CardioSense AI

Automated Phonocardiogram (PCG) Auscultation & Murmur Detection System

System Documentation & Technical Specifications
Model Backbone: CardioResNet-18 (Deep 2D-CNN)
Dataset Benchmark: PhysioNet CinC PCG Challenge
Document Version: 1.0 • Comprehensive Review

1 Project Objective & Clinical Problem Statement

> **Primary Goal:** Provide an automated, high-precision acoustic screening platform for acoustic stethoscope recordings to reliably detect **pathological heart murmurs**, **valvular regurgitation** (e.g., Mitral Regurgitation), and **stenotic ejection turbulence** (e.g., Aortic Stenosis) prior to secondary echocardiogram referral.

- **Clinical Challenge:** Traditional cardiac stethoscope auscultation is highly subjective. Subtle or early-stage murmurs are frequently missed in high-noise clinic environments, leading to delayed cardiac interventions.
- **System Solution:** CardioSense implements a deterministic pipeline: raw acoustic ingestion → zero-phase 4th-order Butterworth DSP filtering → 128-band Log-Mel Spectrogram extraction → PyTorch CardioResNet-18 deep inference → Youden-calibrated optimal thresholding → safety-guardrailed clinical report generation.

2 End-to-End System Architecture Flow

1. AUDIO INPUT — Mic / Upload / Benchmark WAV, MP3, WebM
→
2. DSP CLEANER — 2,000 Hz Resample Butterworth 20-200Hz
→
3. SEGMENTATION — 4.0s Window, 2.0s Hop 50% Temporal Overlap
→
4. MEL-SPECTROGRAM — 128 Mel Bands (1, 128, 63) Tensor
→
5. RESNET-18 CNN — 8 Residual Blocks Skip Connections
→
6. DECISION & REPORT — Threshold $\tau^* = 0.4900$ Guardrailed Summary

3 Audio Ingestion & Resampling Protocol

- **Internal DSP Sample Rate:** Standardized to **2,000 Hz** ($f_s = 2,000\text{ Hz}$).
  - *Nyquist Limit:* $f_{Nyquist} = 1,000\text{ Hz}$ ($f_s / 2$). Fundamental heart valve thuds (S_1, S_2) span 20–150 Hz; pathological murmurs span up to 400 Hz. A 2,000 Hz rate captures 100% of the relevant acoustic envelope without wasting computational memory on unnecessary high frequencies.
- **Channel Downmix:** Multi-channel recordings are downmixed to single-channel mono by averaging: $y_{mono}[n] = \frac{1}{C}\sum_{c=1}^C y_c[n]$.
- **Universal Browser Playback Resampling:** Raw PhysioNet recordings are sampled at 2,000 Hz float64, which native browser decoders (Chrome/Edge) reject with `NotSupportedError`. The backend server automatically resamples audio on-the-fly to **44.1 kHz 16-bit PCM WAV** via `scipy.signal.resample_poly` (39ms execution), enabling instant, crackle-free browser playback.

4 Digital Signal Processing (DSP) & Bandpass Filtering

Filter Implementation: 4th-Order Butterworth Bandpass IIR Filter via `scipy.signal.butter` & `scipy.signal.filtfilt`.

- **Lowcut Frequency (f_{low}): 20.0 Hz** — Strips sensor friction against skin, thoracic breathing movement, muscle tremors, and baseline drift.
- **Highcut Frequency (f_{high}): 200.0 Hz** — Strips environmental room chatter, clinician voices, equipment noise, and 50/60 Hz mains electrical hum.
- **Zero-Phase Filtering (`filtfilt`):** Applies the filter forward and reverse. Result: **Zero phase distortion ($\Delta \phi = 0$)**. Crucial for clinical medicine because causal filters delay high frequencies, distorting the exact time alignment of S_1 (mitral/tricuspid) and S_2 (aortic/pulmonic) valve closures.
- **Peak Amplitude Normalization:** Scales signal amplitude to $[-1.0, 1.0]$: $y_{norm}[n] = \frac{y[n]}{\max(|y|) + 10^{-6}}$.

5 Temporal Segmentation: Why 4.0-Second Windows with 50% Overlap?

CardioSense splits filtered recordings into **4.0-second sliding windows** (8,000 samples) with a **2.0-second hop** (50% overlap):

- **Physiological Verification (4–6 Complete Cardiac Cycles):** At a resting adult heart rate of 60–100 BPM, one cardiac cycle ($S_1 \rightarrow \text{systole} \rightarrow S_2 \rightarrow \text{diastole}$) lasts 0.6–1.0s. A 4.0-second window guarantees 4 to 6 full beats. Shorter windows (e.g. 1.0s) risk catching only half a beat, making rhythm and continuous turbulence indistinguishable.
- **Temporal Murmur Localization:** Slicing allows the system to pinpoint exactly when pathology occurs (e.g., "*Segment 3 (4.0s–8.0s) flagged: elevated turbulence*"), rather than providing only a blind overall label.
- **Prevents Murmur Dilution:** If a patient has a transient 4-second murmur in a 35-second recording, evaluating the whole recording at once dilutes the murmur energy. Slicing isolates the anomaly with high signal-to-noise ratio.
- **50% Boundary Overlap (2.0s Hop):** Prevents murmurs that occur right at a boundary cut (e.g. at second 3.9) from being bisected and missed. Every beat appears fully centered in at least one window.
- **Uniform Matrix Dimensions:** Produces uniform 2D tensors of shape **`(1, 128, 63)`** for fast batch parallel inference.

6 Feature Extraction: 128-Band Log-Mel Spectrogram

Converts 1D acoustic sound waves into 2D time-frequency spectrogram heatmaps via Librosa:

Parameter	Configuration	Technical & Clinical Function
Mel Bands (n_{mels})	**128 bands**	Non-linear perceptual frequency scale emphasizing critical lower-frequency heart valve ranges.
FFT Window (N_{FFT})	**512 samples (256 ms)**	Optimal balance between frequency resolution and temporal localization of valve clicks.
Hop Length (H)	**128 samples (64 ms)**	75% frame overlap providing smooth, continuous time-frequency representation.
Frequency Range	**20.0 Hz – 400.0 Hz**	Broadened beyond 200 Hz to capture high-frequency regurgitant jet murmur harmonics.
Scaling & Range	**Decibel (dB) → [-1.0, 1.0]**	Logarithmic power conversion normalized for numerical stability in neural network layers.

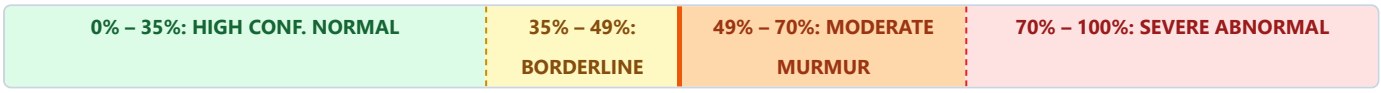
7 Deep Learning Architecture: CardioResNet-18

The primary inference backbone is **CardioResNet-18**, an 18-layer deep residual convolutional neural network:

- **Stem:** 7×7 Conv2D (64 filters, stride 2) → BatchNorm → ReLU → 3×3 MaxPool (stride 2).
- **4 Residual Stages (8 Residual Blocks):**
  - Stage 1: 64 channels (stride 1) • Stage 2: 128 channels (stride 2 downsampling)
  - Stage 3: 256 channels (stride 2 downsampling) • Stage 4: 512 channels (stride 2 downsampling)
- **Residual Basic Block:** $\text{Output} = \text{ReLU}\Big(\text{Conv}_2(\text{ReLU}(\text{Conv}_1(x))) + \text{Shortcut}(x)\Big)$. Identity skip connections allow gradients to flow directly without vanishing, preserving delicate acoustic murmur signatures.
- **Global Pooling & Regularization:** `AdaptiveAvgPool2d((1, 1))` collapses the spatial dimensions, enabling the model to handle recordings of variable length without resizing artifacts. Regularized with **Dropout ($p = 0.35$)** and Batch Normalization.
- **Classifier:** Fully-connected layer mapping 512 features to 2 class logits (Normal vs. Abnormal).

8 Optimal Decision Threshold ($\tau^* = 0.4900$) & Score Ranges

The network outputs an Abnormal Probability Score (P) between 0.00 and 1.00. In clinical medicine, a false negative (missing a diseased heart) is catastrophic. Using **Youden's J Statistic** ($J = \text{Sensitivity} + \text{Specificity} - 1$), the decision threshold was calibrated to **$\tau^* = 0.4900$** (49.0%):



Probability Range	Diagnosis	Clinical Acoustic Profile	Action & Validation Benchmark
0.00 – 0.35 (0%–35%)	NORMAL	Pristine S_1/S_2 closures; quiet, silent baseline; regular rhythm.	Routine preventive checkup. *Benchmark: `a0007.wav` (34.1% abnormal, 65.9% confidence).*
0.35 – 0.49 (35%–49%)	NORMAL	Minor background noise or benign physiologic S_2 splitting. Below threshold.	No immediate referral needed. *Benchmark: `a0004.wav` (33.3% abnormal).*
0.49 – 0.70 (49%–70%)	ABNORMAL	Subtle systolic turbulence or regurgitation jet passing $\tau^* = 0.4900$.	Stethoscope re-auscultation + consider Echocardiogram. *Benchmark: `a0001.wav` (54.4% abnormal).*
0.70 – 1.00 (70%–100%)	ABNORMAL	Severe continuous high-frequency harsh systolic rushing noise.	Urgent Echocardiogram (TTE) referral. *Benchmarks: `a0006.wav` (92.1%), `a0003.wav` (82.4%).*

> **PhysioNet Held-Out Test Set Performance: ROC-AUC: 0.8041 • Sensitivity (Recall on Murmurs): 82.78%** (125/151 abnormal hearts correctly caught) • **Specificity: 63.33%** (95/150 healthy hearts confirmed).

9 Clinical AI Report Agent & Medical Safety Guardrails

Every inference result is processed through an automated safety and governance layer before presentation:

- **Anti-Overclaiming Filter:** Regular expressions replace definitive diagnostic claims:
  - "*You have aortic stenosis*" → "*Acoustic patterns indicate potential signs of aortic stenosis*".
  - "*Confirmed case of heart failure*" → "*Presents acoustic characteristics consistent with...*".
- **Hallucination / Grounding Check:**
  - If $P(\text{Abnormal}) < 0.35$, words like "*critical murmur*" or "*severe pathology*" are strictly suppressed.
  - If $P(\text{Abnormal}) > 0.65$, terms like "*completely healthy heart*" or "*zero abnormalities*" are prohibited.
- **Mandatory Investigational Disclaimer:** Every generated summary appends a statutory reminder: "*CardioSense is an automated acoustic screening and decision-support tool. It does not provide a definitive diagnosis. All findings must be clinically correlated with stethoscope auscultation and echocardiography by a licensed clinician.*"

10 Cryptographic Security & Audit Logging

To guarantee medical audit compliance, reproducibility, and chain-of-custody:

- **SHA-256 Audio Signature:** Upon upload, a 256-bit cryptographic hash of raw audio bytes is generated.
- **SQLite Persistence (`cardiosense.db`):**
  - `AnalysisRecord`: Stores session ID (`CS-XXXXXXXXXXXX`), patient tag, confidence, duration, and findings.
  - `AuditLog`: Records model version, filter frequencies, raw segment probability arrays, and guardrail audit flags.

11 Master Parameter & Configuration Reference Table

System Layer	Parameter Name	Configured Value	Clinical & Engineering Purpose
DSP Ingestion	Sample Rate (f_s)	**2,000 Hz**	Covers 20–400 Hz cardiac sound band with 1,000 Hz Nyquist ceiling.
DSP Ingestion	Channels	**Mono (1-channel)**	Downmixes stereo channels to single acoustic phonocardiogram.
DSP Filter	Bandpass Frequencies	**20.0 Hz – 200.0 Hz**	Strips sensor motion (<20Hz) and ambient speech/hum (>200Hz).
DSP Filter	Filter Type & Order	**Butterworth 4th-Order**	Maximally flat passband with rapid −80 dB/decade attenuation.
DSP Filter	Filtering Algorithm	**Zero-Phase (filtfilt)**	Eliminates phase distortion ($\Delta \phi = 0$); preserves S1/S2 timing.
Segmentation	Window Duration	**4.0 seconds** (8,000 samples)	Captures 4–6 consecutive heart cycles at normal 60–100 BPM.
Segmentation	Hop Duration	**2.0 seconds** (50% overlap)	Prevents murmurs at boundary cuts from being bisected and lost.
Spectrogram	Mel Filter Bands (n_{mels})	**128 bands**	High spectral resolution across auditory non-linear scale.
Spectrogram	FFT Window & Hop	**$N_{FFT}=512$, $H=128$**	256ms time window with 75% frame overlap.
Spectrogram	Frequency Range	**20.0 Hz – 400.0 Hz**	Captures low-frequency valve closures plus high-frequency murmur jet.
Deep Learning	Architecture	**CardioResNet-18**	8 residual blocks with skip connections preventing gradient vanishing.
Deep Learning	Regularization	**Dropout (0.35) + BatchNorm**	Prevents overfitting to specific acoustic recording hardware.
Decision Threshold	Optimal Cut-Off (τ^*)	**0.4900 (49.0%)**	Maximizes Youden's J statistic; prioritizes murmur recall (82.78%).
Browser Playback	Audio Resampling	**44.1 kHz 16-bit PCM WAV**	Universal HTML5 audio decoding eliminating browser playback errors.

12 Benchmark Audio Verification Samples (PhysioNet Cohort)

Sample File	Ground Truth	Model Output	Abnormal Prob.	Acoustic Characteristics
`a0007.wav`	Normal	**NORMAL**	**34.1%** (65.9% Conf.)	Pristine physiological S1/S2 rhythm, clean silent systolic interval, 72 BPM cadence.
`a0006.wav`	Abnormal	**ABNORMAL**	**92.1%** (92.1% Conf.)	Severe pathological murmur; continuous high-amplitude systolic turbulence.
`a0001.wav`	Abnormal	**ABNORMAL**	**54.4%** (54.4% Conf.)	Mitral Regurgitation jet; demonstrates sensitivity right above the $\tau^* = 0.4900$ threshold.